

Module 11: Advanced PostgreSQL

Overview

This module covers the features that distinguish PostgreSQL from other relational databases and from basic SQL usage. Every topic is production-relevant: engineers at companies like Shopify, GitLab, Stripe, and Notion rely on these features for their core data infrastructure. Completing this module unlocks the ability to design sophisticated PostgreSQL-native solutions that outperform comparable application-level implementations.

Prerequisites

- Strong SQL foundations (Modules 1–4)
- PostgreSQL Core familiarity (Module 5)
- Database Design (Module 6)
- Indexes and Query Optimization (Modules 7–8)
- Transactions and Concurrency (Module 9)
- PostgreSQL Internals (Module 10)

Module Contents

File	Topic	Key Concepts
01_jsonb_mastery.md	JSONB deep dive	Operators, GIN indexes, JSONPath, jsonb_set
02_full_text_search.md	Full-text search	tsvector, tsquery, GIN/GiST, ranking
03_partitioning.md	Table partitioning	Range, list, hash; partition pruning; pg_partman
04_materialized_views.md	Materialized views	REFRESH CONCURRENTLY, incremental refresh, patterns
05_extensions_overview.md	Extensions ecosystem	pg_stat_statements, uuid-ossdp, PostGIS, TimescaleDB
06_logical_decoding.md	Logical decoding	Replication slots, pg_logical, wal2json, Debezium
07_foreign_data_wrappers.md	Foreign Data Wrappers	postgres_fdw, file_fdw, import foreign schema
08_listen_notify.md	LISTEN / NOTIFY	Async messaging, pubsub patterns, real-time apps
09_stored_procedures_functions.md	Stored procedures & functions	PL/pgSQL, volatility, SECURITY DEFINER, cursors
10_triggers_and_rules.md	Triggers and rules	BEFORE/AFTER/INSTEAD OF, audit patterns, row versioning

Learning Path

Foundation: 01 (JSONB) → 02 (FTS) → 03 (Partitioning)
Intermediate: 04 (MatViews) → 05 (Extensions) → 09 (Functions)
Advanced: 06 (Logical Decoding) → 07 (FDW) → 08 (Listen/Notify) → 10 (Triggers)

Module Architecture at a Glance

PostgreSQL Advanced Feature Stack

QUERY LAYER	JSONB operators	Full-text search	FDW queries
COMPUTE LAYER	Stored functions	Triggers	Rules
STORAGE LAYER	Partitioned tables	Materialized views	TOAST / JSONB
CHANGE CAPTURE	Logical decoding	LISTEN/NOTIFY	Audit triggers

Quick Reference: Feature Selection Guide

Use Case	Feature
Semi-structured data, flexible schemas	JSONB
Search by keywords with ranking	Full-text search
Very large tables (time series, logs, events)	Partitioning
Pre-computed expensive aggregations	Materialized views
Streaming changes to Kafka / Debezium	Logical decoding
Querying remote databases transparently	Foreign Data Wrappers
Real-time push notifications from DB	LISTEN / NOTIFY
Reusable business logic in the DB	Stored functions
Automatic DML side effects	Triggers
Audit logging and row versioning	Triggers

Key Skills Gained

After completing this module you will be able to:

- Design schemas that blend relational and document (JSONB) columns effectively
- Build full-text search with PostgreSQL without an external search engine
- Partition multi-billion-row tables for query pruning and maintenance efficiency
- Implement audit logging, optimistic locking, and soft-delete using triggers
- Stream database changes to external systems via logical decoding
- Use FDWs to query Postgres, CSV files, and other data sources in a single SQL statement
- Build real-time notification systems using LISTEN/NOTIFY

Related Modules

- **Module 12: Production PostgreSQL** — Operating these features in production (monitoring, tuning)
- **Module 13: Replication & HA** — Logical decoding is the foundation of logical replication
- **Module 15: Backup & Recovery** — Logical decoding slots must be managed to avoid WAL accumulation
- **Module 16: PostgreSQL for Data Engineering** — FDWs, triggers, and logical decoding power ETL pipelines